

DiffGeo notes

SUNAY JOSHI

July 3, 2026

Professor Dafermos: global theorems in Riemannian geometry are typically of the form: “a local geometric assumption and a weak topological assumption combine to yield a strong global geometric property.” In Lorentzian geometry, things are the other way around...

Given ONB $\{e_i\}$ at p , $Ric_p(x, y) = \frac{1}{n-1} \sum (x, e_i, y, e_i)$ in do Carmo convention. (Other authors omit the $\frac{1}{n-1}$.) Equivalently, Ricci is the trace of linear map $z \mapsto Ric(x, z)y$. In particular $Ric_p(v) = \frac{1}{n-1} \sum (v, e_i, v, e_i)$. Recall $K(x, y) = \frac{(x, y, x, y)}{|x \wedge y|^2}$ is the sectional curvature of $\sigma = \text{span}\{x, y\}$, i.e. the gauss curvature of $\exp(\sigma)$ as an embedded submanifold. So Ricci is an average of sectional curvatures. It is clear that Ric is a 2-tensor. In coordinates x_i , Ric can be obtained by contracting the second and fourth components of the Riemann curvature 4-tensor: $Ric_{ik} = \sum_{j,l} R_{ijkl} g^{jl}$.

§1 Hadamard

Complete + nonpositive sectional curvature + s.c. implies diffeo to $\mathbb{R}^n$. Further, the diffeo is given by the exponential map $\exp_p : T_p M \rightarrow M$ (which in this case is defined on all of $T_p M$).

add...

§2 Model spaces

Professor Dafermos: “There is no end to the study of S^n , $\mathbb{H}^n$, and their various quotients. A good geometer should be able to navigate these with ease.”

Quotient by the action of a group G of isometries that act properly discontinuously (else imagine quotienting $\mathbb{R}^2$ by group generated by reflection about the x -axis)...

Quotients of S^n : S^n (s.c. if $n \geq 2$), double covers RP^n (orientable if n odd)... (When $n = 3$, things are interesting: S^3 is homeo to $SU(2)$ and RP^3 is homeo to $SO(3)$.)

Quotients of $\mathbb{R}^n$: $\mathbb{R}^n$, if we quotient by the action of a lattice Γ by translations, then we obtain a torus $\mathbb{R}^n/\Gamma$, the fundamental domain of a Γ -periodic function.

Quotients of $\mathbb{H}^n$: when $n = 2$, using the upper halfplane model, if we quotient by the action of $PSL_2(\mathbb{Z})$ by Mobius transformations, then we obtain the fundamental domain of a modular form (think an Eisenstein series defined by an integer lattice)...

add...

§3 Bonnet-Myers

Bonnet-Myers: Suppose (M, g) is a complete Riemannian manifold. Suppose $Ric_p(v) \geq \frac{1}{r^2} > 0$ for all p, v with $|v| = 1$. Then $diam(M) \leq \pi r$. Hence M compact (?).

(Think sphere: all sectional curvatures are $\frac{1}{r^2}$, so $Ric \equiv \frac{1}{r^2}$. Diameter of the sphere is πr .)

We want to show that for all p, q , there exists a curve γ joining p to q with length $\leq \pi r$.

Since M is complete, by Hopf-Rinow there exists a length-minimizing geodesic γ connecting p to q . Let $\gamma : [0, 1] \rightarrow M$ have length ℓ . We claim $\ell \leq \pi r$. We apply the “second derivative test” to the *energy* functional, which has a local minimum at γ . This reads $E''(0) \geq 0$ for any variation of γ .

The trick is to construct $n - 1$ variations, each moving in an independent direction.

Parallel transport an ON frame $\{e_i(t)\}$ along γ , where e_1 is chosen parallel to γ' , i.e. $e_1(t) = \frac{1}{\ell}\gamma'(t)$. (Recall the speed is ℓ .)

For each j in $2, \dots, n$, we construct a variation in the direction e_j . Let $V_j(t) = \sin(\pi t)e_j(t)$ be the variational vector field (which when integrated yields the actual variation). This is a proper variation because $V_j(0) = V_j(1) = 0$; the sine wave makes a half-period. Recall if γ is a geodesic, then any proper variation is automatically a critical point of the energy: $E'(0) = 0$. So it makes sense to compute $E''(0)$ via the second variation formula. For a proper variation, the quantity $E''(0)$ can be expressed in terms of the index form as

$$E''_j(0) = -2 \int_0^1 (\langle V'', V \rangle - (\gamma', V, \gamma', V)) dt \quad (1)$$

Here we use the shorthand V'' for the second covariant derivative $\frac{D^2 V}{dt^2}$. (This can be checked with a lot of tedious calculation.) We see the sectional curvature of $\text{span}\{\gamma', V\}$ sitting inside the integrand. If we average over j , a Ricci term should pop out.

But we must first massage the integrand. Note $V_j = \sin(\pi t)e_j$. Since e_j is parallel, $V_j'' = -\pi^2 V_j$, so that $\langle V_j'', V_j \rangle = -\pi^2 \sin^2(\pi t)$. Next, since $\gamma' = \ell e_1$, multilinearity of R yields $(\gamma', V, \gamma', V) = \ell^2 \sin^2(\pi t)(e_1, e_j, e_1, e_j)$.

Now we average over j in $2, \dots, n$. Since by definition $Ric_p(e_1) = \frac{1}{n-1} \sum (e_1, e_j, e_1, e_j)$, we obtain

$$\frac{1}{n-1} \sum E''_j(0) = -2 \int_0^1 (-\pi^2 \sin^2(\pi t) - \ell^2 \sin^2(\pi t) \cdot Ric_{\gamma(t)}(e_1)) dt \quad (2)$$

Since γ is length-minimizing, the RHS is nonpositive, so that

$$\int_0^1 (\pi^2 \sin^2(\pi t) - \ell^2 \sin^2(\pi t) \cdot Ric_{\gamma(t)}(e_1)) dt \geq 0 \quad (3)$$

We isolate ℓ , the quantity of interest. Beware: $Ric_{\gamma(t)}(e_1(t))$ depends on t . But luckily, it is bounded below by $\frac{1}{r^2}$. So

$$\pi^2 r^2 = \frac{\int_0^1 \pi^2 \sin^2(\pi t) dt}{\int_0^1 \frac{1}{r^2} \sin^2(\pi t) dt} \geq \ell^2 \quad (4)$$

as desired. ■

So the length bound is really coming out of the inequality

$$L(\gamma)^2 \leq \frac{\int_0^1 \pi^2 \sin^2(\pi t) dt}{\int_0^1 \text{Ric}(e_1) \sin^2(\pi t) dt} \quad (5)$$

for any length-minimizing geodesic γ . The function $f(t) = \sin(\pi t)$ is specially chosen. Maybe others work too.

How can we remember the second variation formula?

$$E''(0) = -2 \int_0^1 (\langle V'', V \rangle - \langle \gamma', V, \gamma', V \rangle) dt \quad (6)$$

Let's consider the Euclidean setting, where the second term is absent and where covariant derivatives are vanilla derivatives. Let $\gamma : [0, 1] \rightarrow \mathbb{R}^n$ be a geodesic (i.e. straight line) and V a proper variational vector field. By definition $E(s) = \int_0^1 \langle \partial_t \gamma, \partial_t \gamma \rangle$ (abusing the notation " γ ")

Differentiate: $E'(s) = \int_0^1 2 \langle \partial_s \partial_t \gamma, \partial_t \gamma \rangle$

Commute partials without picking up curvature, then do product rule trick:

$$E'(s) = \int_0^1 2(\partial_t \langle \partial_s \gamma, \partial_t \gamma \rangle - \langle \partial_s \gamma, \partial_t^2 \gamma \rangle)$$

$$\text{FTC} + \text{proper: } -\frac{1}{2}E'(s) = \int_0^1 \langle \partial_s \gamma, \partial_t^2 \gamma \rangle$$

Differentiate once more and evaluate at $s = 0$ (i.e. our original γ): $-\frac{1}{2}E''(0) = \int_0^1 \langle \partial_s^2 \gamma, \partial_t^2 \gamma \rangle + \int_0^1 \langle \partial_s \gamma, \partial_s \partial_t^2 \gamma \rangle$

The first term is $\langle V', \gamma'' \rangle = 0$ since γ is a geodesic. Commuting partials without picking up curvature, the second term is $\langle V, V'' \rangle$, so that $-\frac{1}{2}E''(0) = \int_0^1 \langle V, V'' \rangle$. By IBP, we also have $\frac{1}{2}E''(0) = \int_0^1 |V'|^2$.

To include the curvature terms, use the relation $\partial_t \partial_s - \partial_s \partial_t = R(\partial_s, \partial_t)$, which comes from the definition of the Riemann curvature tensor. (Here each partial should be interpreted properly.)

§4 Sygne-Weinstein

Our goal is the following relation between positive sectional curvature and simply-connectedness/orientability.

Corollary (Synge's theorem): if M is compact Riemannian manifold with positive sectional curvature, then

- If n is even and M is orientable, then M is simply-connected
- If n is odd, then M is orientable

The theorem statement is a bit clunky, since it arrives at different conclusions based on the parity of n . Let's consider some examples.

S^n : when n even (think S^2), Synge implies that S^n is simply-connected. However, when n odd,

we can't guarantee simply-connected; RP^3 has fundamental group $\mathbb{Z}/2$, as it is double covered by S^3 .

RP^n : recall that this is the space of lines through the origin in $\mathbb{R}^{n+1}$, or equivalently the result of identifying antipodal points on S^n . Recall that RP^2 is non-orientable (think of a gluing diagram). Synge implies that RP^3 is orientable (in fact, RP^n for any odd n).

How do we prove Synge? Well, at least in the even case, it seems useful to study the universal cover $\tilde{M}$ of M . If we can show the group of deck transformations is trivial, then M is simply-connected.

Recall that $\tilde{M}$ is defined purely topologically. A natural question is whether $\tilde{M}$ admits a differentiable structure. The answer is yes. There exists a "Riemannian covering metric" $\tilde{g}$ on $\tilde{M}$ such that the covering map $\pi : \tilde{M} \rightarrow M$ is a local isometry. It's defined in the obvious way, by pulling back g by π . Thus all completeness/curvature assumptions on (M, g) lift to $(\tilde{M}, \tilde{g})$.

A diffeo f of $\tilde{M}$ is a deck transformation iff $\pi \circ f = \pi$. (This is with respect to M , of course.) These deck transformations are isometries?... It turns out that the group of deck transformations is isomorphic to $\pi_1(M)$. (Think of $\pi : \mathbb{R}^2 \rightarrow \mathbb{T}^2$...)

To prove Synge's theorem when n is even, we want to study the deck transformations of $(\tilde{M}, \tilde{g})$. If $\pi_1(M) \neq 0$, then there exists a deck transformation $\phi \neq e$ without fixed points...

We therefore ask when a manifold with positive sectional curvature has an isometry with no fixed points...

Synge-Weinstein: Let (M, g) be a compact Riemannian manifold with positive sectional curvature. Let f be an isometry of M . If n even, assume f preserves orientation. If n odd, assume f reverses orientation. Then f has a fixed point.

Let's see how Synge-Weinstein implies Synge. Since M is compact, $K \geq \varepsilon > 0$ for some constant ε . Then $(\tilde{M}, \tilde{g})$ is complete and inherits $K \geq \varepsilon$. By Bonnet-Myers, $(\tilde{M}, \tilde{g})$ is compact. Orientability... If n even, we can apply Synge-Weinstein to $(\tilde{M}, \tilde{g})$ to deduce that all isometries of $\tilde{M}$ have fixed points. Thus there cannot exist any non-identity deck transformations of $\tilde{M}$. Equivalently, $\pi_1(M) = 0$.

n odd...

Read backwards, Synge tells us what metrics we can put on a given topological space. There is no metric on $\mathbb{T}^2$ with positive sectional curvature. You can ask the same about $S^2 \times S^2$.

§5 Rauch comparison

Index lemma (only one trick), Rauch comparison, length comparison (duality), Sphere theorem..

§6 Classical Lie groups

add...

§7 Curvature dimension conditions

Ricci lower bound equivalent to Bakry-Emery log-Sobolev inequality (Sara discussed in the case of the generator being the Laplacian), rate of convergence of Markov process to equilibrium measure (i.e. the law μ_t of a solution to SDE converges to some μ), analogy with discrete Markov chains, mixing time, MCMC, Langevin MC requires further discretization in time (which must be done carefully), Hamiltonian MC...

§8 Wasserstein geometry

From discussion with Dwip, restrict to submanifold $\mathcal{P}_2$, the W_2 distance on the manifold allows us to interpolate between μ and ν via a geodesic, geodesic convexity, metric?...